

Group Project Assignment 1: Societal Problem Definition

1. One recent newspaper article that describes a societal challenge

This news article was written by Richard Partington, senior economics correspondent, and published on 16-08-2026 in The Guardian, which is a UK-based national daily newspaper known for independent journalism and in-depth economic and business coverage

<https://www.theguardian.com/business/2026/aug/16/new-uk-cost-of-living-crisis-looms-rising-energy-bills-inflation> (screenshots at the end of the document)

2. The Challenge Identified in the Article

At its core, the article points to a fresh wave of financial pain for everyday UK households—a renewed cost-of-living squeeze driven by a spike in household energy bills and stubborn global inflation. With Ofgem raising the energy price cap by 13%, utility costs are shooting back up, threatening to push national inflation toward 3%. To make matters worse, global tensions like conflict in the Middle East are shaking up energy markets, driving prices up even higher. The result is a double whammy for ordinary families: their money doesn't stretch as far, and potential interest rate hikes from central banks means borrowing costs and mortgages could become even harder to manage.

- **Why are you interested in this topic?**

What makes this issue so compelling is how clearly it connects giant, abstract economic forces to the quiet decisions people has to make around their kitchen tables. Its a vivid look at how a crisis thousands of miles away can instantly ripple down to a family's monthly budget. Beyond that, it highlights a high-stakes balancing act for policymakers—watching governments and central banks try to tame inflation without completely crushing the financial stability of average citizens are both a critical economic challenge and a deeply human one.

- **For whom is this challenge relevant (who are your stakeholders)?**

This issue hits closest to home for everyday families, particularly working- and middle-class households who feels every extra pound on an energy bill or grocery receipt. Beyond individual consumers, its top of mind for government leaders and central bankers at the Bank of England, who has to figure out how to cool off inflation without causing a recession or leaving vulnerable families behind. Finally, energy suppliers, regulators, and small business owners are key players here too, as they try to keep their doors open while juggling volatile wholesale energy prices and employees who desperately needs higher wages just to keep up.

3. Per team member, find, select, and read a different scientific peer-reviewed article on this topic.

Member 1

This article is crucial for understanding the team's selected challenge because it provides empirical, model-based evidence for how sudden energy price shocks affect household economics on a systemic scale. While the news coverage describes real-time events in the UK such as rising consumer price index inflation following Ofgem price cap hikes this study models the underlying mechanisms driving that crisis. It demonstrates how wholesale energy cost increases filter directly through supply chains into everyday consumer goods, acting as a regressive burden that hits lower-income households the hardest due to the high proportion of their income spent on energy and basic utilities.

Guan, Y., Yan, J., Shan, Y., Zhou, Y., Heng, Y., Wang, R., Meng, J., Zhang, Q., & Hubacek, K. (2023). Burden of the global energy price crisis on households. *Nature Energy*, 8(3), 304–316. <https://doi.org/10.1038/s41560-023-01209-8>

Regarding the article subtype, this paper is classified as an Original Research Article that uses quantitative modeling through global multi-regional input–output analysis.

Member 2

Valenti, D., Bazzana, F., & Gabbriellini, R. (2023). Energy price shocks, exchange rates and inflation nexus. *Energy Economics*, 128, 107156. <https://doi.org/10.1016/j.eneco.2023.107156>

This empirical study investigates how international energy price shocks transmit directly into domestic inflation rates and exchange rate volatility. It is highly relevant to our chosen societal challenge because it provides econometric evidence on how sudden energy cost spikes rapidly push up headline consumer prices, illustrating the systemic vulnerability of economies to commodity price swings and the difficulty of containing cost-push inflation.

Regarding the article subtype, this is an original research article because the authors run their own statistical and econometric models on real price data rather than just summarizing other papers.

Member 3

This article is relevant to our challenge because it shows how the same driver behind our news article (energy price shocks) doesn't hit everyone the same way. The Guardian article talks about how the UK energy price cap went up 13%, which is pushing inflation close to 2.9%. The PNAS study looks at 31 developed countries and finds that elderly and low-income households carry a much bigger energy cost burden than younger or wealthier groups, both from their direct energy use (heating, electricity) and from the energy costs hidden in the products and services they buy. This helped us understand that inflation caused by energy prices isn't a uniform shock, it clearly hits some groups harder than others, which is important for figuring out who the vulnerable stakeholders are in our challenge.

Tian, P., Feng, K., Sun, L., Hubacek, K., Malerba, D., Zhong, H., Zheng, H., Li, D., Zhang, N., & Li, J. (2024). Higher total energy costs strain the elderly, especially low-income, across 31 developed countries. *Proceedings of the National Academy of Sciences*, 121(12), e2306771121. <https://doi.org/10.1073/pnas.2306771121>

In terms of the subtype, this is an original research article. The authors used their own model (a global multi-regional input-output model) together with real household expenditure survey data from 31 countries to measure energy cost burdens by age and income group, so it's actual original analysis, not a review or an opinion piece.

Member 4

The article describes the underlying history of the UK's inflation dynamics while also providing a similar case of energy crisis caused by an external energy shock due to UK's reliance on external energy imports. Furthermore, it compares how UK, USA and Germany dealt with consequences of COVID-19, Brexit and energy price shocks of 2021-2024. Article further describes the causes and mechanisms of cost-push inflation that is currently affecting the UK's energy market. It also shows that the UK remains vulnerable to external energy price shocks, as increases in energy costs can spread into higher production, transport and household costs across the wider economy. Finally, showing the limitations of monetary policy, since higher interest rates can reduce demand but cannot directly solve the underlying problem of limited or expensive energy supply.

Sarwar MF and Abatti AL and Khalid N, 2026. Inflation dynamics and economic impacts in the UK, USA, and Germany (2015–2024): a comparative macroeconomic analysis. *Sci Soc Insights*, 5: 58-65. <https://doi.org/10.65822/j.sasi/2026.007>

New UK cost of living crisis looms with soaring energy bills forecast to lift inflation

Latest reading on Wednesday likely to frame scale of challenge for new PM to ease pressures on households amid Iran war fallout



📷 Economists said a rise in inflation was likely after the energy regulator lifted its cap on household gas and electricity bills in July. Photograph: Jacob King/PA

British households are facing a renewed cost of living squeeze, with official figures expected to show this week that soaring energy bills drove up inflation in July to close to 3%.

As the Iran war continues to send shock waves through global energy markets, economists predict the surge in UK gas and electricity bills last month will push Britain's headline inflation rate to 2.9%.

In a fresh squeeze on household budgets, the figures from the [Office for National Statistics](#) (ONS) due on Wednesday are forecast to show a jump from a rate of 2.6% in June.

The [Bank of England](#) is considering raising interest rates from as early as

September in response to fears over stubbornly high inflation becoming entrenched in the economy.

Reports also emerged overnight that the water regulator Ofwat was considering plans to impose “surge pricing” on water usage during droughts. It could see customers charged more for water use during summer and less in winter, or else charged higher prices when they pass a threshold of water use, according to the Daily Telegraph.

The latest snapshot will highlight the challenge facing Andy Burnham’s government to ease the financial pressure on households and businesses before a difficult autumn budget.

Economists said a rise in inflation as measured by the consumer prices index was likely after Ofgem, the energy regulator, lifted its cap on household gas and electricity bills by 13% in July.

Thomas Pugh, the chief economist at the accountancy firm RSM UK, said the increase would add about 0.44 percentage points to headline inflation. This would be partly offset by a fall in petrol and diesel prices.

“The cost of living squeeze is set to return to the headlines,” he said. “[Higher inflation is] adding fresh pressure to household budgets and complicating the outlook for interest rates.”

As the stop-start Middle East war fuels volatility in the global oil price, countries around the world are facing renewed inflationary pressures and heightened uncertainty over the scale of the economic hit.

Britain’s economy has shown **more resilience than initially feared**, with official figures last week showing that it continued to grow in the first half of 2026 at the fastest pace in the G7. Inflation also **fell by more than expected in June, to 2.6%** - down from a peak of 3.8% last year.



📷 It is hoped the fall in petrol and diesel prices could partly offset the rise in energy bills.
Photograph: Andy Rain/EPA

Inflation had been on track to fall to close to 2% before the outbreak of the Iran war.

However, economists warn the impact from the conflict is likely to weigh more heavily in the second half of the year after the increase in the Ofgem energy price cap.

Separate figures on the UK jobs market due to be released on Tuesday are expected to show a continued slowdown in wage growth.

With households under pressure from the renewed rise in energy costs, the [Bank of England](#) predicts UK inflation will reach 3.2% before the end of the year, despite government measures to limit the impact.

Burnham used his first week as prime minister to announce a series of “breathing space” measures to ease the cost of living, including [cutting VAT](#) to reduce consumer electricity bills by an average of £45 a year from October.

The Bank said it expected the policy, alongside a £2 cap on bus fares in England, to lower the headline inflation rate by 0.1 percentage point.

However, Threadneedle Street [kept borrowing costs unchanged](#) last month as it warned that a worst-case scenario - involving further escalation in the Middle East war - could drive inflation to a peak of 4.5% by the middle of 2027.

City investors anticipate two quarter-point interest rate rises from the Bank before the end of next year, with financial markets giving an almost one-in-four chance of the first increase in the base rate, now 3.75%, coming at its next policy meeting in September.

Victoria Scholar, the head of investment at the financial platform Interactive Investor, said: “Inflation is expected to continue to rise, peaking above 3% later this year, as the UK economy continues to grapple with the backdrop of elevated energy prices and the effective gridlock in the strait of Hormuz.

“The Bank is likely to carry out roughly one 25 basis point hike by the end of the year as it looks to temper the risk of overheating and help push the inflation rate back in the direction of [its] 2% target.”

